Vaishnavi Kashyap

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Data engineer with experience building production pipelines from normalized schema design and ETL through data quality enforcement, serving ops teams and executives across GCP and AWS. M.S. Computer Science, GPA 4.0, UMass Amherst, May 2026. [portfolio]

EXPERIENCE

Software Engineer — **Nov 2023 – Aug 2024**
Saint Gobain — *Mumbai, India*

- Replaced a patchwork of factory spreadsheets across 5+ sites with a single source of truth by designing a normalized PostgreSQL schema (central emissions fact table linked to employees, categories, and submissions), Python ETL scripts transforming 10–12 sustainability metrics per employee, SQL aggregation queries powering real-time Scope 1/3 KPI dashboards for **300+ users**, and S3 signed-URL exports for secure data distribution.
- Ensured data quality and reliability end-to-end via DRF serializer validation, schema-level constraints, and automated cross-category reconciliation before downstream aggregation; sustained **99.9% uptime** via PostgreSQL index tuning and AWS CloudWatch pipeline monitoring.
- Built versioned DRF REST APIs for data ingestion and retrieval with structured error responses, serving as the data contract between factory-floor data producers and downstream reporting consumers.
- Delivered Angular + Chart.js self-serve analytics dashboards for ops teams and site managers, reducing ad-hoc reporting requests and giving non-technical stakeholders direct visibility into emissions trends; GitHub Actions CI/CD; Agile/Scrum.

Data Science Intern — **Jun 2022 – Aug 2022**
CloudSufi — *San Francisco*

- Owned the full ETL pipeline and ARIMA forecasting model on **GCP Vertex AI** across **1M+ records** of Scope 1 emissions enriched with weather covariates; applied feature engineering, holdout validation with residual analysis, and deployed to production with BigQuery-connected AppSheet dashboards for non-technical ops teams.
- Cut analyst query time by **70%** by designing BigQuery SQL views with multi-table joins, rolling 2-week aggregation windows, and data quality checks surfacing **1M+ daily data points**; presented findings to the CEO and full ML team.

PROJECTS

Stock Market ELT Pipeline with Real-Time Streaming | Airflow, dbt, Kafka, PostgreSQL, Docker [github] — **Apr 2026 – Jun 2026**

- Prevented bad data from ever reaching the warehouse by gating every downstream run behind automated validation across a 3-layer dbt warehouse (raw → staging → marts) unifying a batch ELT pipeline and a real-time Kafka streaming layer for **10 tracked tickers**; a **16-task Airflow DAG** with parallel-isolated fetch tasks and automatic retries lands data in **under 5 minutes**.
- Guaranteed safe re-runs and zero data loss on streaming failures by pairing **25+ automated tests across 5 dbt models** with idempotent upserts and a **10-partition Kafka producer/consumer layer** with dead-letter-queue handling, all containerized across an **11+ service Docker stack**.

Real-Time Air Quality Monitoring Platform | FastAPI, PostgreSQL, Celery [github] — **Feb 2026 – May 2026**

- Prevented duplicate writes and pipeline stalls in a live OpenAQ air-quality ingestion pipeline by decoupling ingestion into async Celery + Redis tasks backed by an idempotent PostgreSQL time-series store (`INSERT ON CONFLICT DO NOTHING`) with ingestion-time data quality checks; verified with **27 pytest unit tests** and Docker CI/CD.
- Resolved a **74% v1 error rate** by moving synchronous weather API calls into a Celery-populated DB table; the refactored pipeline **load-tested to 230 RPS at 100 concurrent clients with 0 errors**.

PUBLICATIONS

Interferometric Coherence for SAR Land-Cover Classification | *IEEE IGARSS 2022* [paper]
As ML lead, built a CNN pipeline for 6-class land-cover classification over Sentinel-1 SAR imagery and integrated interferometric coherence as an engineered feature, improving overall accuracy by **11%** and urban and vegetation class accuracy by **15–25%**.

EDUCATION

University of Massachusetts Amherst — **May 2026**
M.S. in Computer Science | GPA: 4.0 / 4.0 Coursework: Data Science Fundamentals, Quantitative Analysis, Applied Statistics, Software Engineering

Vellore Institute of Technology, Vellore — **May 2023**
B.Tech. in Computer Science | GPA: 3.86 / 4.0 Coursework: Data Structures & Algorithms, Operating Systems, Database Systems

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, Bash/Shell
Data Engineering: ETL/ELT Pipelines, Airflow, dbt, Schema Design, Data Quality, Pipeline Monitoring, Pandas, NumPy
Databases & Stores: PostgreSQL, BigQuery, SQLAlchemy, Alembic, Celery, Redis, Kafka
Cloud & Infra: GCP (Vertex AI, BigQuery), AWS (S3, EC2, CloudWatch), Docker, GitHub Actions, CI/CD, Git
Backend & APIs: FastAPI, Django, DRF, Node.js, REST API design
Visualization: Chart.js, AppSheet
Practices: Agile/Scrum, unit testing (pytest), data validation, stakeholder communication, documentation